Homework Bridge

My dad always knew how to do my math homework. He just couldn't get it to me.

AI disclosure: Claude (Anthropic) wrote the code and helped polish this write-up. I directed the project — the idea, the research into how other countries teach math, the design decisions, and the conversation with my dad are mine.

Live demo: `https://homework-bridge-rho.vercel.app` **Code:** `https://github.com/Adarsh-Samanu/homework-bridge`

The problem

My dad grew up in India and learned all of this math there — long division, subtraction with borrowing. He is good at it. He speaks Telugu, and English is hard for him. I never learned enough Telugu for him to just explain it to me instead.

So when I got stuck on a worksheet in elementary and middle school, this is what it looked like: I'd be sitting there with the paper, my dad right next to me, and he *knew the answer* — and there was no way for it to get from him to me. He couldn't read the sheet. I couldn't follow his explanation. One language apart, at the same table.

I asked him about it while building this. He said something like this would have really helped him back then — he wanted to teach me the way he had learned it, and couldn't.

And translating the worksheet would not have fixed us. Hand him my homework in perfect Telugu and he still can't help, because it says things like *"use a number bond to decompose the second number."* There is no Telugu word for **number bond** — not because it's a hard translation, but because the thing doesn't exist in what he learned. He was taught column subtraction with borrowing. He'd be staring at a diagram he had never seen, now in Telugu.

It isn't only vocabulary. **Long division sits somewhere else on the page** in Mexico, Vietnam, and Brazil. `3,5` **means three and a half** in most of Latin America, so a parent can read `3.5` as thirty-five and never find out. `1,00,000` **is one lakh** in India. `187 r 1` — nothing says the `r` means remainder.

TalkingPoints, ParentSquare, ClassDojo and Google Lens all translate school communication well. Every one would have left my dad exactly where he was.

The solution

Photograph the worksheet or paste it in, pick your language and the country you went to school in. Then, in your language:

1. **What the homework is actually asking** — plainly, not word-for-word
2. **The way the school wants it done**, step by step
3. **The same problem done the way *you* were taught** ← nothing else does this
4. **How the two line up**, and the one place they really differ
5. **Notation warnings** — the comma and decimal traps above
6. **Three questions to ask your kid** that check understanding without giving the answer

Worked examples always use **different numbers** than the real homework, so you can't copy them onto the sheet. This helps a parent help; it doesn't do the homework.

Try the third sample — Telugu, India, number bonds. That is my dad's exact situation.

Tech stack

Front end & hosting. Next.js 16 (App Router), React 19, TypeScript, deployed on Vercel. Read-aloud uses the browser's Web Speech API — free and offline. No database, no accounts, nothing stored.

Models, through Featherless AI's OpenAI-compatible API: `GLM-5.2` does all reasoning and writes the explanation in the parent's language; `Qwen3-VL` transcribes photos and is never allowed to solve anything. An Anthropic-SDK adapter is swappable via one environment variable. Every stage walks a fallback chain, because individual models go busy without warning.

The part that matters most is a data file. Asked directly, the model gave vague and inconsistent answers about how other countries teach arithmetic. So we researched the real differences and wrote them down country by country in `lib/methods.ts`. Deciding that file had to exist is the call that makes the comparison trustworthy.

Correctness is machine-checked. The first working version told a Telugu-speaking parent that $62 - 27 = 55$ — confidently, in the panel a parent is meant to trust. A homework helper that gets numbers wrong is worse than none, because now the parent looks wrong in front of their kid. So reading and reasoning were split apart, and `eval/run.py` now re-computes every equation the model writes and fails the run on any mismatch. The three demo worksheets pass: 8, 16 and 8 equations checked, zero wrong.

What's next

- **More countries.** Six are covered. The Philippines, Haiti, Nigeria and Central America are the obvious gaps.
- **Handwriting.** Printed worksheets read reliably; handwritten ones are the weakest part.
- **Test with real families,** not just mine. One interview shaped this; ten would shape it better.
- **Speed.** Answers take 8–20 seconds live. The demo worksheets are pre-rendered because of it.
- **Beyond math** — science and reading have their own untranslatable methods.